[INCH-POUND]
MIL-H-16384E
19 October 1988
SUPERSEDING
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12 October 1978
(See 6.10)

MILITARY SPECIFICATION

HYDROMETER SETS AND PARTS, SYRINGE, LEAD-ACID STORAGE BATTERY TESTING

This specification is approved for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers hydrometer sets and replacement parts for use in measuring the specific gravity of electrolyte in submarine, portable, radio, and aircraft lead-acid storage batteries and for measuring mixing acid used for electrolyte in such batteries.

1.2 Hydrometer sets shall be of the following types as specified (see 6.2). Types A and B shall be furnished as complete sets or as individual parts (see 3.5 and 6.5).

Type A - For submarine lead-acid storage batteries.
Type B - For portable and radio lead-acid storage batteries
Type C - For aircraft lead-acid storage batteries
Type D - For measuring mixing acid

2 APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, Naval Sea Systems Command, SEA 5523, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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SPECIFICATIONS

FEDERAL

GG-T-258 - Tester, Battery Electrolyte Solution
PPP-F-320 - Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes

MILITARY

MIL-P-116 - Preservation, Methods of
MIL-T-3618 - Thermometer, Self-Indicating, Bimetallic
MIL-L-19140 - Lumber and Plywood, Fire-Retardant Treated

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-2073-1 - DoD Materiel Procedures for Development and Application of Packaging Requirements

(Unless otherwise indicated, copies of federal and Military specifications and standards are available from the Naval Publications and Forms Center, (ATTN NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099)

2.1.2 Other Government publications. The following other Government publication forms a part of this document to the extent specified herein Unless otherwise specified, the issue is that cited in the solicitation

PUBLICATION

MARE ISLAND NAVAL SHIPYARD
Procedure CAK-173 - For Checking the Accuracy of Hydrometers Used in the Maintenance/Operation of Lead-Acid Storage Batteries.

(Copies of Procedure CAK-173 are available from Mare Island Naval Shipyard, Vallejo, CA 94592)

2.2 Non-Government publications The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
D 3951 - Standard Practice for Commercial Packaging (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103)

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(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article When specified (see 6.2), prior to beginning production a sample shall be subjected to first article inspection (see 6.4) in accordance with 4 3

3.2 Material Material shall be as specified herein (see 6.3).

3 2.1 Rubber. Soft rubber bulbs, plugs, and tubes shall be made of rubber compound, properly vulcanized, and containing a minimum of 25 percent, by volume, crude rubber and a total rubber hydrocarbon content of at least 60 percent Free sulfur shall not exceed 0 5 percent The mineral fillers shall be free from all substances likely to have a deleterious effect on the rubber compound The rubber articles shall be of such size that they will fit as applicable, with airtight connections. The slight ridge left by the parting of the mold will not be considered objectionable The rubber shall be free of abrasions, blemishes, blooms, breaks, deformation or pits.

3.2.2 Glass. All glass parts shall be of high grade transparent flint glass, or equivalent, free from blisters, spots, chips, checks, scratches, bubbles, or defects of any nature Glass tubing shall be of uniform diameter and wall thickness throughout, and the ends shall be carefully finished.

3.2 3 Recovered materials Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified

3 3 Operation The hydrometer shall measure the specific gravity of storage battery electrolyte referred to water at 60 degrees Fahrenheit (°F) temperature through the scale ranges and with the accuracy specified herein The accuracy shall not be impaired by contact, friction, or adhesion between the hydrometer and the syringe barrel

3 4 Construction.

3 4 1 Hydrometer The hydrometer shall consist of a scale enclosed in a ballasted airtight glass tube float The lower end of the tube shall form the float bulb, and shall enclose ballast material in a fixed position The upper

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end of the tube shall form the float stem, and shall enclose a scale in a fixed position Both bulb and stem shall be of practically uniform diameter and circularity of section, and there shall be no deviation from axial alignment apparent to the eye The hydrometer shall be dry inside, perfectly sealed, and shall float in a vertical position in a solution of the highest specific gravity indicated on the scale. Hydrometers shall have projections to prevent undue contact with the barrel and yet permit free circulation of electrolyte. They shall be calibrated in accordance with Mare Island Procedure CAK-173 to provide the accuracy specified herein. Projections shall be rounded and symmetrically spaced, at least six shall be used around the float bulb and shall not extend more than 3/32 inch beyond the float body.

3 4.1.1 Hydrometer scale The hydrometer scale shall be neatly printed on high-grade paper and shall be easily readable and firmly fixed in position. Figures shall be plain and as large, as practicable, the scale shall be within the limiting length and specific gravity range specified herein. The scale shall carry a mark indicating each minor division to specific gravity change with figures at the mark indicating every fifth minor division All figures shall be even multiples of 0.005. The scale for type A hydrometers shall be triple, for types B and D double, and for type C, either double or single Examples of single, double, and triple scales are shown on Figure 1 (which is provided for information only).

3 4 1.2 Hydrometer labeling Each hydrometer float shall be labeled with the manufacturer's name or trademark, a unique serial number for the manufacturer identifying the hydrometer, and with the temperature standardization of the hydrometer The hydrometer temperature standardization shall be given as the temperature at which the hydrometer will read the electrolyte's specific gravity correctly, over the temperature of the reference liquid. For example· a hydrometer that is designed to read correctly at 80°F compared to a gas-free distilled water reference liquid at 60°F shall be indicated as an 80/60 type hydrometer

3 4 2 Syringe barrel The syringe barrel shall be a straight hollow glass cylinder of practically uniform thickness throughout the entire length. The ends of the barrel shall be provided with rolled edges or beads made by pressing the glass while hot.

3.4.3 Rubber bulb. The rubber bulb shall be in accordance with 3.2.1, shall be not less than 3/32 nor more than 11/64 inch thick for all types, and shall be of such a size and construction that, when pressed between the thumb and two fingers until the inside walls touch and then released, sufficient electrol· yte shall be drawn into the barrel to permit accurate reading of the hydrometer (or float), even though the electrolyte has a specific gravity equal to the lowest value on the appropriate hydrometer scale as specified herein The bulb shall fit over the bead or rolled edge at the top end of the barrel so as to prevent any air leakage

3 4 4 Top plug The top of the syringe barrel shall be tightly fitted with a perforated soft rubber plug to prevent the hydrometer from entering the rubber bulb

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3 5 Detail

3 5 1 Type A The type A hydrometers shall be furnished as complete sets or individual parts, as specified (see 6.2) A complete type A set shall consist of the following parts:

(a) Bulb
(b) Top plug
(c) Bottom plug and tube (in one piece)
(d) Syringe barrel
(e) Hydrometer float scale 1.228 to 1.316
(f) Hydrometer float scale 1 060 to 1.240
(g) Bimetallic thermometers

3 5 1.1 Dimensions The parts shall conform to the dimensions specified in table I

TABLE I Dimensions

Dimensions	Maximum	Minimum
	Inches	Inches
Syringe barrel·		
Length	12.063	11 938
Inside diameter	0 766	0.725
Outside diameter	.884	.797
Thickness	.059	.039
Hydrometer syringe-bottom plug and rubber tube		
Length	7.063	6 938
Square flange or hexagonal side	1 250	1 250
Hydrometer float·		
Length	7.563	----
Diameter over projections	0 720	----
Scale length	----	3

3.5.1.2 Float accuracy The float shall be accurate within plus or minus one minor scale division throughout the scale range specified in 3 5.1

3 5 1.3 Thermometers Thermometers shall be bimetallic self-indicating dial thermometers in accordance with MIL-T-3618.

3.5.1.4 Bottom plug and tube The bottom of the syringe barrel shall be fitted with a one-piece soft rubber plug, with a rubber tube projecting therefrom The plug and tube shall be provided with a hole or slot of sufficient size to allow liquid to enter the barrel The hole or slot shall be so located that the hydrometer float will not close the opening when the liquid is forced out of the barrel The plug shall be so fitted in the barrel that it will not leak or

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fall out of place if it deteriorates. The plug shall have a square or hexagonal flange projection outside the syringe barrel which will act as a buffer and will prevent the hydrometer syringe from rolling when placed on a flat surface

3.5.2 Type B The type B hydrometers shall be furnished as complete sets or individual parts, as specified (see 6.2). A complete type B set shall consist of the following parts

(a) Bulb
(b) Top plug
(c) Bottom plug and soft rubber tube (in one piece)
(d) Bottom plug and hard rubber tube (two piece)
(e) Syringe barrels
(f) Hydrometers, float scale 1.100 to 1.300
(g) Bimetallic thermometer

3.5.2.1 Dimensions The parts shall conform to the dimensions specified in table II

TABLE II Dimensions.

Dimensions	Maximum Inches	Minimum Inches
Syringe barrel		
Length	7 063	6 938
Inside diameter	0 640	0 593
Outside diameter	758	671
Thickness	059	039
Hydrometer syringe-bottom plug tube		
Soft tube length	3	2 938
Hard tube length (in addition to part inserted in plug)	4 500	4 438
Tube, hard or soft outside diameter	0.312	0 250
Square flange side	1	938
or		
Hexagonal flange side	1 250	1 250
Hydrometer float.		
Length	4.50	----
Diameter over projections	0 570	----
Scale length	----	1 5

3.5.2.2 Float accuracy. The float shall be accurate within plus or minus one minor scale division throughout the scale range specified

3.5.2.3 Thermometers. The thermometers shall be bimetallic self-indicating dial thermometers in accordance with MIL-T-3618

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3 5 2 4 Bottom plug and soft rubber end table. The one-piece plug bottom end tube shall be as specified in 3 5 1.4 and shall have the dimensions specified in table II.

3.5.2.5 Bottom plug and hard rubber (two piece) end tube. The two-piece plug bottom end tube shall consist of a plug, which shall be so fitted in the barrel that it will not leak or fall out of place if it deteriorates. The rubber plug shall have a square or hexagonal flange projection outside the syringe barrel which will act as a buffer and will prevent the hydrometer syringe from rolling when placed on a flat surface. The hard rubber tube shall have the dimensions specified in table II, and shall be inserted into the soft rubber plug bottom and be firmly attached

3.5.3 Type C The type C hydrometers shall indicate the lowest value of specific gravity marked on the scale with 20 cubic centimeters of electrolyte within the syringe The type C hydrometer shall conform to class 1, styles C and D of GG-T-258 and as specified herein A complete type C set shall consist of the following parts

(a) One bulb
(b) One top plug
(c) One bottom plug and tube (in one piece)
(d) One syringe barrel
(e) One hydrometer, scale 1.160 to 1.250
(f) One hydrometer, scale 1.210 to 1.300
(g) One thermometer and correction scale

3 5 3 1 Syringe barrel dimensions The syringe barrel shall not exceed 8-11/16 inches in length The maximum wall thickness shall be 0 059 inch and the minimum 0 039 inch.

3 5.3 2 Scale dimensions. The hydrometer scale shall be not less than 1-1/2 inches long The hydrometer scale shall be either double or single, and shall be visible from any direction.

3.5.3.3 Float accuracy. The float shall be accurate within plus or minus 0 005 specific gravity at 80°F throughout the scale range specified in 3 5 3(e) or (f)

3.5.3 4 Thermometers. A readily readable thermometer tube mounted on a calibrated correction scale shall form an integral part of the set. It shall be so arranged that the thermometer will be directly actuated by the electrolyte within the syringe The correction scale shall include corrections to be added to or subtracted from the hydrometer float readings for electrolyte temperatures ranging from 0 to 120°F over a minimum length of 1-1/4 inch scale This scale shall be marked with the corrections for each 3 degree change in temperature. Corrections to be added and those to be subtracted shall be indicated in different colors. Corrections shall be further identified by the wording "Add to float reading" and "Subtract from float reading", or other wording approved by the Command or agency concerned The temperature correction scale shall be based on the zero correction, to be 60°F This scale shall be based on a correction of 0 001 specific gravity for every 3°F

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3.5.3.4.1 Thermometer mounting. Thermometers and correction scales shall be mounted on and protected by the molded bottom plug and tube, or shall be mounted in any other position on the unit where they are easily readable and protected, provided such alternate mounting is specifically approved by the Command or agency concerned.

3 5 3.5 Bottom plug and tube. The bottom of the barrel shall be fitted with a one piece soft rubber plug and tube. The plug and tube piece shall be designed to protect the assembled hydrometer set from breakage or rolling when it is placed on a flat surface. It may also be designed to house the thermometer and correction scale

3 5 3 5 1 Plug end The plug end shall include a groove to receive the bead or rolled edge of the barrel, and shall be so fitted on the barrel that it will not leak at any time or fall out of place should it deteriorate in service. A slot or hole of sufficient size to permit liquid to enter the barrel shall be provided, and shall be so located as to prevent its obstruction by the hydrometer when liquid is being forced out of the barrel The plug or thermometer housing, if used, shall act as a buffer to the float and prevent its accidental breakage

3 5 3 5 2 Tube end The tube, for sampling electrolyte by insertion through battery-cell cover openings, shall be straight, at least 2-1/4 inches long, not more than 3/8-inch outside diameter, and shall either be molded integral with the plug can or vulcanized into place on the plug. The end of the tube shall be made of soft rubber so that the separators will not be damaged

3.5.4 Type D. for measuring mixing acid.

3 5 4 1 Scale The hydrometer scale shall be from 1 150 to 1 400 and shall be not less than 1-1/2 inches long

3.5 4.2 Dimensions The hydrometer shall have a maximum diameter over projections of 0 57 inch and a maximum length of 5 inches.

3 5 4 3 Float accuracy The float shall be accurate within plus or minus 0 005 specific gravity throughout the scale range specified in 3 5 4 1

3 6 Resistance to sulfuric acid (rubber bulbs, plugs, and tubes) The finished bulbs, plugs and tubes shall be resistant to sulfuric acid to the extent that when immersed in water white acid of 1.300 specific gravity for 96 hours, there shall be no roughening of the sample and no sediment

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements

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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5 The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program (see 6 3) The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections The inspection requirements specified herein are classified as follows

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection Prior to production, first article inspection shall be performed on one complete set, or replacement parts, as applicable, when a first article sample is required (see 3.1 and 6.2). This inspection shall include the examination of 4.5 and the tests of 4.6

4 4 Quality conformance inspection Quality conformance inspection shall include the examination of 4 5 and the test of 4 6 2

4.5 Visual and dimensional examination. The complete sets or, replacement parts shall be visually and dimensionally examined to determine conformance with this specification

4 6 Tests

4.6 1 Acid resistance Sample rubber parts (see 6.6.3) shall be immersed in 1 300 specific gravity water white acid for 96 hours to determine conformance with 3 2.1.

4 6 2 Calibration Each hydrometer float shall be tested for calibration in accordance with Mare Island Procedure CAK-173.

4 7 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging or preparation for delivery requirements of referenced documents listed in section 2, see 6 7.)

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5.1 General.

5 1.1 Navy fire-retardant requirements

(a) Treated lumber and plywood. When specified (see 6.2), all lumber and plywood including laminated veneer material used in shipping container and pallet construction, members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows.

Levels A and B - Type II - weather resistant.
Category 1 - general use
Level C - Type I - non-weather resistant
Category 1 - general use

(b) Fiberboard Fiberboard used in the construction of class-domestic, non-weather resistant fiberboard and cleated fiberboard boxes including interior packing forms shall meet the flamespread index and the specific optic density requirements of PPP-F-320 and amendments thereto.

(c) Cushioning and wrapping materials The use of excelsior, newspaper, shredded paper (all types), and similar hygroscopic or nonneutral materials and all types of loose fill materials for packaging (preservation and packing) applications such as cushioning, fill, stuffing, and dunnage is prohibited Materials selected for cushioning and wrapping shall have properties (characteristics) for resistance to fire (see 6 8). Cushioning or wrapping materials, as applicable, shall be provided to prevent item and package damage and to prevent free movement of the container contents

5 2 Preservation Preservation shall be level A, C or commercial as specified (see 6 2)

5 2 1 Level A

5.2.1.1 Type A and B hydrometer sets. Type A and B hydrometer sets contained in storage cases (see 3.5.1 and 3.5.2) shall be unit protected in accordance with method III of MIL-P-116 Each storage case shall be unit packed in a weather resistant, fiberboard box in conformance with 5.2.1 3.

5.2.1.2 Type C and D hydrometer sets. Type C hydrometer sets, component parts (see 3.5.3), shall have the individual parts unit protected in accordance with method III of MIL-P-116. All parts comprising a single set shall be cushioned and unit packed in a water resistant folding, set-up or metal edged paperboard or fiberboard box in conformance with 5.2 1 3

5 2.1.3 Unit and intermediate containers Unit and intermediate containers and requirements shall be in accordance with MIL-STD-2073-1, table I, appendix F Container selection and other options such as type, variety, grade and style shall be at the option of the contractor. Box closure shall be in accordance with the applicable container (box) specification or appendix thereto except that weather resistant fiberboard boxes shall be closed in accordance with method V

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5.2.2 Level C. Preservation of hydrometer sets shall be as specified for level A except that the unit and intermediate paperboard containers shall be of the nonweather resistant or domestic construction and fiberboard of the class-domestic/fire-retardant For closure of the fiberboard box, method I using pressure-sensitive adhesive tape shall apply.

5.2.3 Commercial. Commercial packaging (cleaning, preservation, cushioning, and unit pack) requirements shall be in accordance with ASTM D 3951.

5.3 Packing. Packing shall be level A, B, C, or commercial as specified (see 6.2).

5 3 1 General requirements for levels A. B. and C. Containers selected (see 5.3.2) shall be of minimum weight and cube consistant with the protection required, uniform size, and contain hydrometer sets of one type (see 1 2)

5.3 2 Level A. B. and C containers. Hydrometer sets preserved as specified (see 5.2) shall be packed in exterior shipping containers for the level of packing specified (see 5 3), in accordance with table VII, exterior shipping container requirements, of MIL-STD-2073-1, appendix C, and herein Unless otherwise specified (see 6 2), container selection and options shall be at the contractor's option

5.3.2 1 Caseliners. closure and gross weight.

5.3.2.1 1 Caseliners. Unless otherwise specified (see 6 2), level A shipping containers containing hydrometer sets preserved level C or commercial shall be provided with waterproof caseliners in accordance with MIL-STD-2073-1.

5 3 2 1 2 Closure Container closure, reinforcing, or banding shall be in accordance with the applicable container specification or appendix thereto except that weather-resistant fiberboard boxes shall be closed in accordance with method V and reinforced with non-metallic or tape banding and domestic or fire-retardant fiberboard boxes shall be closed in accordance with method I using pressure sensitive tape

5.3 2 1.3 Weight Wood, plywood, and cleated type containers exceeding 200 pounds gross weight shall be modified by the addition of skids in accordance with MIL-STD-2073-1 and the applicable container specification or appendix thereto

5 3 3 Commercial Hydrometer sets preserved as specified (see 5 2), shall be packed for shipment in accordance with ASTM D 3951 and herein.

5.3.3.1 Container modification. Shipping containers exceeding 200 pounds gross weight shall be provided with a minimum of two, 3- by 4-inch nominal wood skids laid flat, or a skid- or sill-type base which will support the material and facilitate handling by mechanical handling equipment during shipment, stowage and storage

5 4 Marking, levels A. B. C. and commercial In addition to any special marking required (see 6.2), interior (unit and intermediate) packs and shipping containers shall be marked including bar coding for shipment, stowage, and storage in accordance with MIL-STD-2073-1, appendix F

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6 NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory)

6 1 Intended use. The hydrometer sets covered by this specification are intended for use in testing electrolyte acid and submarine, portable, radio, and aircraft lead-acid storage batteries.

6 2 Acquisition requirements Acquisition documents must specify the following

(a) Title, number, and date of this specification
(b) Type required (see 1.2).
(c) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2 1 1 and 2.2).
(d) When first article is required (see 3.1)
(e) Whether types A or B are to be supplied as complete sets or as parts (see 3 5.1 and 3.5.2).
(f) When fire-retardant material is required (see 5 1 1(a))
(g) Level of preservation and level of packing required (see 5 2 and 5 3)
(h) Container selection and options if other than the contractor's (see 5.2.1.3 and 5.3.2).
(i) When caseliners are not required (see 5 3.2 1 1)
(j) Special markings required (see 5.4).

6.3 Consideration of data requirements The following data requirements should be considered when this specification is applied on a contract The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/ provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference paragraph	DID number	DID title	Suggested tailoring
3 2	DI-E-2121	Certificate of compliance	----
4 1 1	DI-R-4803	Inspection system program plan	----

The above DID was cleared as of the date of this specification The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423

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6.4 First article When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item should be a preproduction sample a first article sample, a first production item, a sample selected from the first ___ production items, a standard production item from the contractor's current inventory (see 3 1), and the number of items to be tested as specified in 4 3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract Bidders should not submit alternate bids unless specifically requested to do so in the solicitation

6 5 Types A and B. Types A and B are not intended to be stocked as complete sets They should normally be ordered and stocked as individual parts

6 6 Sampling

6 6 1 Lot A lot consists of complete sets, or replacement parts, as applicable, offered for delivery at one time and produced on the same facilities, using identical materials, and manufacturing and assembly procedures.

6 6 2 Sampling for examination Random samples of the hydrometer or replacement parts, as applicable, should be selected from each lot in accordance with inspection level II of MIL-STD-105. Samples should be subjected to both the visual and dimensional examination of 4 5

6 6 3 Sampling for test Random samples of the hydrometer or parts, as applicable, should be selected from each lot in accordance with inspection level S-2 of MIL-STD-105 The samples should be subjected to the test of 4 6 2

6 7 Sub-contracted material and parts. The packaging or preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped

6 8 Cushioning and wrapping materials (see 5.1.1(c)) Materials having properties for resistance to fire and acceptable for use within unit packs and shipping containers for Navy acquisitions are:

Material	Specification
Paper, kraft, treated (fire-resistant)	A-A-1894
Paper, kraft, wrapping	UU-P-268, type II, grade C or D
Fiberboard	PPP-F-320, class - domestic/ fire-retardant
Plastic film, flexible, cellular	PPP-C 795, class 3 - fire-retardant
Polystyrene expanded, resilient	PPP-C 850, grade SE

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Plastic, open cell, cushioning	PPP-C-1842, type I, style B
Bound fiber	PPP-C-1120, type III or IV, class C
Rubber, latex foam	MIL-R-5001, grade A
Rubber, cellular	MIL-R-6130, grade A
Fibrous glass	MIL-C-17435
Polystyrene foam	MIL-P-19644, type II
Rubber, cellular, synthetic	MIL-R-20092, class 5
Polyurethane foam	MIL-P-26514
Polyurethane foam, flexible, open cell	MIL-F-81334
Foam, combustion-retardant, for cushioning supply items aboard Navy ships	MIL-F-87090

6 9 Subject term (key word) listing.

Hydrometer scale
Plug end
Syringe barrel
Thermometer mounting
Tube end

6.10 Changes from previous issue Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

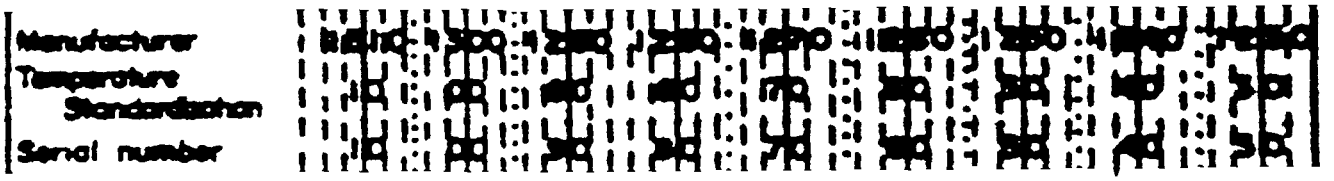
Custodians:
Navy - SH
Air Force - 99

User activities
Army - EL, MI
Navy - YD, MC

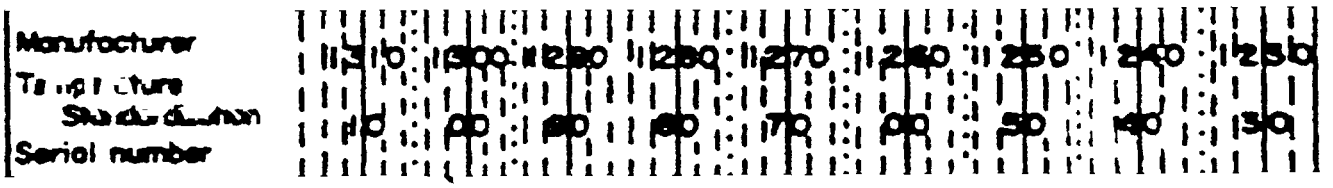
Preparing activity.
Navy - SH
(Project 6630-0477)

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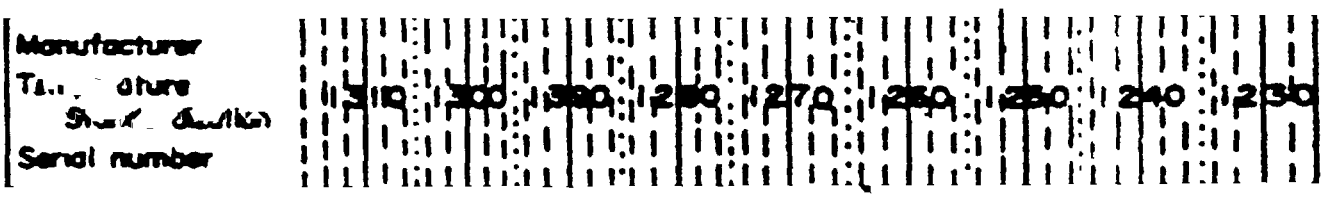
Triple Hydrometer Scale



Double Hydrometer Scale



Single Hydrometer Scale



SH 11310

The 5-point (dotted) graduations are to be in red, all others will be in black (Note This is for information purposes only; it is not an engineering drawing)

FIGURE 1 Hydrometer scales

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER
MIL-H-16384E

2 DOCUMENT TITLE HYDROMETER SETS AND PARTS, SYRINGE, LEAD-ACID STORAGE BATTERY TESTING

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4 TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last, First, MI)* – Optional

b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional

c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional

8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426

PREVIOUS EDITION IS OBSOLETE

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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